

Browser to evidence to adviser

Individual request-flow explanation and submission guide | 6 October 2026

Example question

A registered user asks: "What changes if PUE rises to 1.40?" The browser sends the question and bounded conversation history to POST /api/adviser. The browser never receives the OpenAI or Fingrid key and cannot send an authoritative role or system instruction.

Identity and permission checks

Sites supplies authenticated identity headers. The server looks up the stable identity in D1 users. Anonymous callers receive 401; signed-in but unregistered callers receive 403. The endpoint also checks HTTPS, same-origin requests, JSON size and question length. A D1 reservation enforces per-user and shared request limits; failed requests still count.

Evidence and controlled tools

The server retrieves the current saved design and relevant classified claims, with real source records. It sends those records and explicit instructions to OpenAI Responses. Retrieved text and conversation history are untrusted data. Tools can retrieve approved design/metric/claim records, calculate energy, inspect the deterministic investment model, or read a fixed approved API. They accept neither arbitrary SQL nor an arbitrary URL.

Deterministic arithmetic and return path

For 20 MW IT and PUE 1.40, calculate_energy returns 28 MW and 245.28 GWh at 8,760 hours. That hypothetical calculation does not alter D1. The model returns structured answer/evidence/assumption/calculation/decision/uncertainty sections. The server validates citation identifiers against retrieved records, attaches genuine source URLs and displays warnings for citation problems, preserving the latest project behavior. Warnings are advisory: users must inspect evidence and must not treat every generated statement as verified.

Saved changes and audit

A team administrator may separately save PUE on Manage evidence. The server validates inputs, updates the design, regenerates calculation claims and writes an audit record atomically. Subsequent adviser requests read the updated D1 value. Editors can add reviewed source/claim pairs and refresh only approved APIs. Viewers cannot edit. Source refresh validates the full response before its transaction; failed retrieval retains the last valid data and records a visible failure.

Financing questions

get_investment_analysis uses current saved financial assumptions plus the saved design, returning all nine option/scenario results from application code. It labels them estimates, not quotations. The Investment model page exposes formulas, annual cash flows, assumptions and sensitivity controls. No profit, availability or professional engineering guarantee is produced.

Verification and submission guide

Deliverable map

Website source: `site/`. Model: `/investment`. Country research, demand, engineering, finance and governance: `/decision`. Evidence and timestamps: `/evidence`. Written memo: `investment-memo.docx` (two pages). Physical diagram: `deliverables/system-diagram.pdf` (one page). D1 schema: `d1-schema.sql`. Requirements: `fr-table.md`. Source/API inventory: `d1-schema.md` plus D1 sources. Numerical snapshot: `deliverables/financial-results.json`. Detailed current checks and commands: `deliverables/test-results.md`.

What the checks establish

Pure tests check arithmetic, equal delivered service across options, cash-flow identities, utilization/grid-delay stress, source parsers, tool boundaries and citation warnings. SQLite transaction tests check refresh success and failed-response retention. Local HTTP tests check authentication, viewer/editor/admin restrictions, saved design/finance, evidence insertion, and role-assignment boundaries. A production build and TypeScript checks establish that the source compiles, not that the remote deployment is current.

Production publication dependency

Sites returned “Sites is not yet enabled for this workspace” during this work. The existing live site has not been updated here. In the original Sites-enabled workspace, publish this exact source with appended migration `0003_project_management.sql`, retain server-only keys, and run the same public/authenticated acceptance checks. Never rewrite applied migrations `0000–0002`. Bootstrap the first administrator only through a trusted database operator after verifying that person’s registered stable identity.

Evidence that remains unknown

University commitments, site-specific connection price/date/capacity, hardware and lease guarantees, actual water demand, hourly emissions and integrated backup capability remain unknown. They are decision gates, not fabricated zero values. Financial and engineering planning allowances are explicitly separate from verified data. The model can support an approve/reject/request-more-evidence decision without claiming the site is ready for construction.

Excluded and manual submission items

The five-minute presentation and two-minute demonstration video are excluded at the user’s request. Enter the published URL into the course shared document from an authorized account after deployment. Review the individual explanation above and adapt it to your own understanding before submission.